

Introduction to Classification

Sabyasachi Chatterjee

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1 Classification Problems

In the regression problems that we have studied so far, the response variable was quantitative. For example, we might try to predict a person’s blood pressure, a student’s score on an exam, or the price of a house. In a classification problem, the response variable is qualitative. It takes values in a finite set of classes. For example,

$$Y \in \{\text{spam}, \text{not spam}\},$$

or, in a multiclass problem,

$$Y \in \{\text{setosa}, \text{versicolor}, \text{virginica}\}.$$

The goal is to use a feature vector X to predict the class of Y . Thus, a classifier is a function

$$C(x) \in \mathcal{C},$$

where $\mathcal{C}$ is the set of possible classes. In many statistical and machine learning applications, however, we do not only want the final predicted class. We often want the probabilities of the different classes. For example, it is usually more useful to know that the probability of default is 0.42 than to be told only “default” or “no default.”

Thus, in a binary classification problem, the basic object we want to estimate is

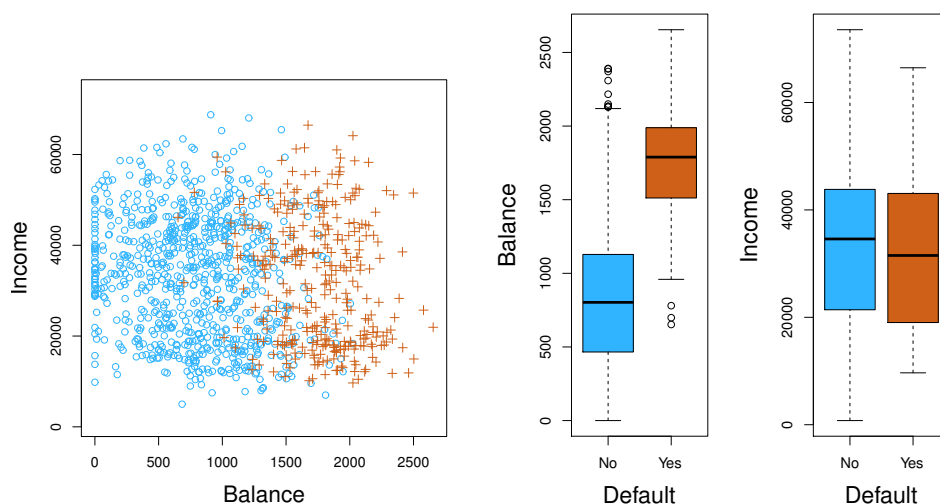
$$P(Y = 1 \mid X = x).$$

Once this probability has been estimated, we can turn it into a classification rule by choosing a threshold.

1.1 Example: Credit Card Default

Let us see a credit card default data set. The response variable is whether a customer defaulted on their credit card debt. The predictors include variables such as balance, income, and student status.

Example: Credit Card Default



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Figure 1: Credit card default data. The plot and boxplots suggest that balance is strongly associated with default, while income is less clearly separated across the two groups.

Figure 1 suggests a natural statistical question. Given a customer's balance and income, can we estimate the probability that the customer will default? In other words, can we estimate

$$P(\text{default} \mid \text{balance}, \text{income})?$$

This is a probability estimation problem. A final yes/no classification can be made afterwards, but the more fundamental object is the conditional probability of default.

1.2 Can We Use Linear Regression?

A first thought is to code the response numerically. For the default data, define

$$Y = \begin{cases} 1 & \text{if the customer defaults,} \\ 0 & \text{if the customer does not default.} \end{cases}$$

Now suppose we fit a linear regression of Y on the predictors. This idea is not as unreasonable as it may first seem. Indeed, because Y only takes the values 0 and 1, we have

$$E(Y \mid X = x) = P(Y = 1 \mid X = x).$$

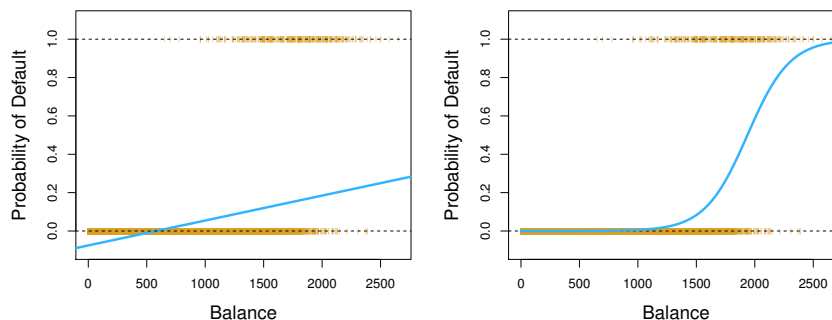
Thus, if we could model the conditional expectation of Y well, we would also be modeling the conditional probability of the positive class.

However, there is a problem. A linear regression model has the form

$$E(Y \mid X = x) = \beta_0 + \beta_1 x_1 + \cdots + \beta_m x_m.$$

The right hand side can be negative or larger than one. Therefore, linear regression can produce fitted values that cannot be interpreted as probabilities.

Linear versus Logistic Regression



The orange marks indicate the response Y , either 0 or 1. Linear regression does not estimate $\Pr(Y = 1|X)$ well. Logistic regression seems well suited to the task.

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Figure 2: Linear regression versus logistic regression for predicting probability of default from balance. The logistic curve stays between zero and one.

The issue is not that linear regression is always a terrible classifier in binary problems. In fact, it can sometimes classify reasonably well. The issue is that it is not an ideal model for probabilities. A probability should always lie between zero and one. Logistic regression fixes this problem by replacing the linear model for the probability with a linear model for the log-odds.

1.3 The Logistic Model

The key idea behind logistic regression is to start with a linear predictor

$$\eta(x) = \beta_0 + \beta_1 x_1 + \cdots + \beta_m x_m,$$

and then transform this linear predictor into a probability.

The transformation used in logistic regression is the logistic function

$$g(z) = \frac{e^z}{1 + e^z}.$$

Remark 1.1. *This seems somewhat arbitrary. We could have presumably used any other strictly increasing function bijectively mapping the real line to $(0, 1)$.*

Thus the logistic regression model is

$$P(Y = 1 \mid X = x) = \frac{\exp(\beta_0 + \beta_1 x_1 + \cdots + \beta_m x_m)}{1 + \exp(\beta_0 + \beta_1 x_1 + \cdots + \beta_m x_m)}.$$

Equivalently, if

$$p(x) = P(Y = 1 \mid X = x),$$

then

$$p(x) = \frac{e^{\eta(x)}}{1 + e^{\eta(x)}}.$$

The advantage of this model is immediate. No matter what value the linear predictor $\eta(x)$ takes, the fitted probability $p(x)$ always lies between zero and one.

For very negative values of the linear predictor, the probability is close to zero. For very positive values, the probability is close to one. In the middle, the curve changes most rapidly.

The logistic model has a particularly simple interpretation in terms of odds. The odds are

$$\frac{p(x)}{1 - p(x)}.$$

For example, if $p(x) = 0.8$, then the odds are

$$\frac{0.8}{0.2} = 4.$$

This means that the event is four times as likely to occur as not to occur.

For the logistic model, taking logarithms gives

$$\log \left(\frac{p(x)}{1 - p(x)} \right) = \beta_0 + \beta_1 x_1 + \cdots + \beta_m x_m.$$

Thus logistic regression is a linear model for the log-odds of the positive class.

This gives a useful interpretation of the coefficients. If x_j increases by one unit while all other variables remain fixed, then the log-odds increase by β_j . Equivalently, the odds are multiplied by e^{β_j} . The usual caveat of interpreting this as an adjusted coefficient (adjusting for the other covariates in the model) remains true here and in any generalized linear model (to be discussed later).

Remark 1.2. *We can fit a nonlinear function to log odds as well; using any nonlinear method we will learn later in the course.*

1.4 Fitting the Logistic Model

The logistic regression model contains an unknown parameter vector

$$\beta = (\beta_0, \beta_1, \dots, \beta_m)^T.$$

These parameters determine how the predictors influence the probability of belonging to the positive class. Once the parameters have been specified, the model assigns a probability

$$P(Y = 1 \mid X = x) = \frac{\exp(\beta_0 + \beta_1 x_1 + \cdots + \beta_m x_m)}{1 + \exp(\beta_0 + \beta_1 x_1 + \cdots + \beta_m x_m)}.$$

The unknown parameters are estimated from data using the method of *maximum likelihood* to obtain $\hat{\beta} = (\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_m)^T$. We shall give a formal description of this procedure in the next section. The fitted model can then be used to estimate probabilities of default, spam, disease, or any other binary outcome of interest.

1.5 A Typical Software Output

After fitting a logistic regression model, statistical software typically produces output such as that shown below.

Maximum Likelihood

We use maximum likelihood to estimate the parameters.

$$\ell(\beta_0, \beta) = \prod_{i:y_i=1} p(x_i) \prod_{i:y_i=0} (1 - p(x_i)).$$

This *likelihood* gives the probability of the observed zeros and ones in the data. We pick β_0 and β_1 to maximize the likelihood of the observed data.

Most statistical packages can fit linear logistic regression models by maximum likelihood. In **R** we use the **glm** function.

	Coefficient	Std. Error	Z-statistic	P-value
Intercept	-10.6513	0.3612	-29.5	< 0.0001
balance	0.0055	0.0002	24.9	< 0.0001

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Figure 3: Logistic regression fit for the credit card default data.

Making Predictions

What is our estimated probability of **default** for someone with a balance of \$1000?

$$\hat{p}(X) = \frac{e^{\hat{\beta}_0 + \hat{\beta}_1 X}}{1 + e^{\hat{\beta}_0 + \hat{\beta}_1 X}} = \frac{e^{-10.6513 + 0.0055 \times 1000}}{1 + e^{-10.6513 + 0.0055 \times 1000}} = 0.006$$

With a balance of \$2000?

$$\hat{p}(X) = \frac{e^{\hat{\beta}_0 + \hat{\beta}_1 X}}{1 + e^{\hat{\beta}_0 + \hat{\beta}_1 X}} = \frac{e^{-10.6513 + 0.0055 \times 2000}}{1 + e^{-10.6513 + 0.0055 \times 2000}} = 0.586$$

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Figure 4: Estimated probabilities of default from the fitted logistic model.

Lets do it again, using **student** as the predictor.

	Coefficient	Std. Error	Z-statistic	P-value
Intercept	-3.5041	0.0707	-49.55	< 0.0001
student[Yes]	0.4049	0.1150	3.52	0.0004

$$\widehat{\Pr}(\text{default}=\text{Yes}|\text{student}=\text{Yes}) = \frac{e^{-3.5041+0.4049 \times 1}}{1 + e^{-3.5041+0.4049 \times 1}} = 0.0431,$$

$$\widehat{\Pr}(\text{default}=\text{Yes}|\text{student}=\text{No}) = \frac{e^{-3.5041+0.4049 \times 0}}{1 + e^{-3.5041+0.4049 \times 0}} = 0.0292.$$

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Figure 5: Estimated probabilities of default from the fitted logistic model.

Logistic regression with several variables

$$\log \left(\frac{p(X)}{1 - p(X)} \right) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$$

$$p(X) = \frac{e^{\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p}}{1 + e^{\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p}}$$

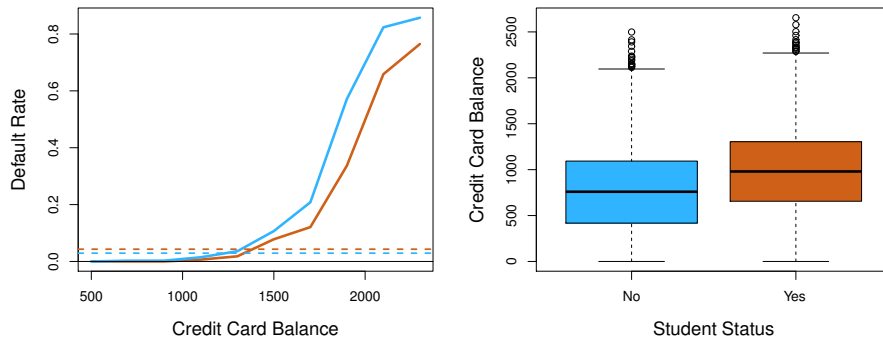
	Coefficient	Std. Error	Z-statistic	P-value
Intercept	-10.8690	0.4923	-22.08	< 0.0001
balance	0.0057	0.0002	24.74	< 0.0001
income	0.0030	0.0082	0.37	0.7115
student[Yes]	-0.6468	0.2362	-2.74	0.0062

Why is coefficient for **student** negative, while it was positive before?

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Figure 6: Estimated probabilities of default from the fitted logistic model.

Confounding



- Students tend to have higher balances than non-students, so their marginal default rate is higher than for non-students.
- But for each level of balance, students default less than non-students.
- Multiple logistic regression can tease this out.

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Figure 7: Estimated probabilities of default from the fitted logistic model.

Remark 1.3. *The above is an interesting example and shows that multiple regression works nicely to bring out the real effect of being a student on defaulting even though being a student is positively correlated with default rate. Again, in a multiple regression model, we should really interpret the coefficients as like univariate coefficients only after regressing out the other predictors in the model.*

The regression table contains quantities that should look familiar from linear regression. For each coefficient we see

- an estimate $\hat{\beta}_j$,
- a standard error,
- a z -statistic,
- and a p -value.

At first sight, these outputs raise several important questions. For example, where do the standard errors come from? Is the z -statistic approximately standard normal? The usual justification for the table is frequentist and is valid asymptotically (roughly when n is large). In the next section we develop the Bayesian inference underlying logistic

regression and show how the quantities reported by software can also be interpreted from this perspective.

1.6 Multiclass Logistic Regression

So far we have discussed binary classification. Logistic regression also has a multiclass version. Suppose we have K classes

$$Y \in \{1, 2, \dots, K\}.$$

We want to estimate all the conditional class probabilities

$$P(Y = 1 \mid X = x), \dots, P(Y = K \mid X = x),$$

which must be nonnegative and must sum to one.

One natural model is the softmax model:

$$P(Y = k \mid X = x) = \frac{\exp(\beta_{0k} + \beta_{1k}x_1 + \dots + \beta_{pk}x_p)}{\sum_{\ell=1}^K \exp(\beta_{0\ell} + \beta_{1\ell}x_1 + \dots + \beta_{p\ell}x_p)}.$$

This is also called multinomial logistic regression. We have a linear function for each class.

Remark 1.4. *There is some redundancy in the parameters here. We would really need $K - 1$ linear functions; however this form seems popular.*

The softmax form guarantees that all probabilities are positive and that they sum to one. The predicted class is often chosen to be the class with the largest estimated probability:

$$\hat{C}(x) = \operatorname{argmax}_{1 \leq k \leq K} \hat{P}(Y = k \mid X = x).$$

This same softmax transformation and multiclass logistic regression often appears in the last layer of modern neural networks. In a neural network classifier, the earlier layers produce features, and the last layer often uses a softmax to turn scores into class probabilities. In a language model, the same idea is used to assign probabilities to possible next tokens. We will see this later in the course.

2 Inference

We are again in the usual regression setting where we observe data

$$(y_i, x_{i1}, x_{i2}, \dots, x_{im}) \quad \text{for } i = 1, \dots, n.$$

There are m explanatory variables $x_1, \dots, x_m$ and one response variable. Suppose now that the response variable is binary, so that

$$y_i \in \{0, 1\}.$$

Let

$$x_i = (1, x_{i1}, \dots, x_{im})^T$$

and

$$\beta = (\beta_0, \beta_1, \dots, \beta_m)^T.$$

Then the model can be written compactly as

$$y_i \stackrel{\text{independent}}{\sim} \text{Bernoulli} \left(\frac{\exp(x_i^T \beta)}{1 + \exp(x_i^T \beta)} \right).$$

2.1 Bayesian Inference

The likelihood is

$$L(\beta) = \prod_{i=1}^n \left(\frac{\exp(x_i^T \beta)}{1 + \exp(x_i^T \beta)} \right)^{y_i} \left(1 - \frac{\exp(x_i^T \beta)}{1 + \exp(x_i^T \beta)} \right)^{1-y_i}.$$

This can be simplified to

$$L(\beta) = \prod_{i=1}^n \frac{\exp(y_i x_i^T \beta)}{1 + \exp(x_i^T \beta)}.$$

The log-likelihood is therefore

$$\ell(\beta) := \sum_{i=1}^n [y_i(x_i^T \beta) - \log(1 + \exp(x_i^T \beta))].$$

Suppose $x_1^T, \dots, x_n^T$ form the rows of the $n \times p$ design matrix X , where $p = m + 1$. The unknown parameters in the logistic regression model are $\beta_0, \dots, \beta_m$. In contrast to the usual linear regression model, there is no additional variance parameter σ^2 .

As in linear regression, we may use the prior

$$\beta_0, \beta_1, \dots, \beta_m \stackrel{\text{i.i.d.}}{\sim} \text{Unif}(-C, C) \quad (1)$$

for a large C . The posterior density of β is then

$$\begin{aligned} f_{\beta|\text{data}}(\beta) &\propto L(\beta) \times \text{prior} \\ &\propto \left[\prod_{i=1}^n \frac{\exp(y_i x_i^T \beta)}{1 + \exp(x_i^T \beta)} \right] I\{\beta_0, \beta_1, \dots, \beta_m \in (-C, C)\} \\ &= \exp(\ell(\beta)) I\{\beta_0, \beta_1, \dots, \beta_m \in (-C, C)\}. \end{aligned}$$

This posterior density is not a familiar standard density. If $p = 1$ or $p = 2$, one can plot it. In higher dimensions, one can use MCMC to sample from it. Here, however, we derive a multivariate normal approximation to the posterior. This is called the Laplace approximation or posterior normal approximation.

2.2 Normal Approximation to the Posterior

To get the normal approximation, first drop the indicator in the prior. For large C , this will typically be irrelevant near the high posterior probability region. Thus

$$f_{\beta|Y=y}(\beta) \propto \exp(\ell(\beta)).$$

The normal approximation is obtained by a second-order Taylor expansion of $\ell(\beta)$ around the MLE $\hat{\beta}$, defined by

$$\hat{\beta} := \operatorname{argmax}_{\beta \in \mathbb{R}^p} \ell(\beta).$$

The gradient of the log-likelihood is

$$\nabla \ell(\beta) = \sum_{i=1}^n \left(y_i - \frac{\exp(x_i^T \beta)}{1 + \exp(x_i^T \beta)} \right) x_i. \quad (2)$$

The Hessian is

$$H\ell(\beta) = - \sum_{i=1}^n \frac{\exp(x_i^T \beta)}{(1 + \exp(x_i^T \beta))^2} x_i x_i^T.$$

Taylor expansion around $\hat{\beta}$ gives

$$\ell(\beta) \approx \ell(\hat{\beta}) + \langle \nabla \ell(\hat{\beta}), \beta - \hat{\beta} \rangle + \frac{1}{2} (\beta - \hat{\beta})^T H\ell(\hat{\beta}) (\beta - \hat{\beta}).$$

Because $\nabla \ell(\hat{\beta}) = 0$, and because $\ell(\hat{\beta})$ is constant in β , we get

$$\begin{aligned} f_{\beta|Y=y}(\beta) &\propto \exp \left(\frac{1}{2} (\beta - \hat{\beta})^T H\ell(\hat{\beta}) (\beta - \hat{\beta}) \right) \\ &= \exp \left(-\frac{1}{2} (\beta - \hat{\beta})^T \left[-H\ell(\hat{\beta}) \right] (\beta - \hat{\beta}) \right). \end{aligned}$$

This is the unnormalized density of a multivariate normal distribution with mean $\hat{\beta}$ and covariance matrix $[-H\ell(\hat{\beta})]^{-1}$.

Let W be the $n \times n$ diagonal matrix with diagonal entries

$$W_{ii} = \frac{\exp(x_i^T \hat{\beta})}{(1 + \exp(x_i^T \hat{\beta}))^2}.$$

Then

$$-H\ell(\hat{\beta}) = X^T W X. \quad (3)$$

Therefore the posterior normal approximation is

$$\beta \mid Y = y \approx N\left(\hat{\beta}, (X^T W X)^{-1}\right). \quad (4)$$

The standard errors of $\hat{\beta}_0, \dots, \hat{\beta}_m$ are the square roots of the diagonal entries of $(X^T W X)^{-1}$.

It turns out that Bayesian inference based on this normal posterior approximation coincides with the usual large-sample frequentist inference for logistic regression. Thus the standard errors, confidence intervals, and z -tests reported by statistical software can be viewed through this Bayesian normal approximation.

2.2.1 Frequentist Inference

For frequentist inference, one needs to understand the sampling distribution of the maximum likelihood estimator $\hat{\beta}$. Unlike ordinary linear regression, the logistic regression MLE does not have a closed-form expression. Consequently, its sampling distribution cannot be derived exactly in finite samples.

Instead, one relies on asymptotic theory for maximum likelihood estimators. Under suitable regularity conditions,

$$\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} N(0, I(\beta)^{-1}),$$

where

$$I(\beta) = \lim_{n \rightarrow \infty} \frac{1}{n} X^T W X$$

is the Fisher information matrix. Equivalently,

$$\hat{\beta} \approx N(\beta, (X^T W X)^{-1}),$$

for large sample sizes. The asymptotic normality of maximum likelihood estimators is a general result (uses Taylor expansions as well); see van der Vaart [4]. Thus the standard errors reported by statistical software are obtained from the diagonal entries of $(X^T W X)^{-1}$, and confidence intervals and hypothesis tests are based on this asymptotic normal approximation.

Notice that this is a fundamentally frequentist justification and is asymptotic in nature. It is also exactly the same covariance matrix that appeared in the Bayesian normal approximation derived in the previous section. Thus Bayesian (with normal approximation) and frequentist inference lead to similar conclusion, atleast when the sample size is large.

2.3 Newton's Algorithm for Computing the MLE

The maximum likelihood estimator $\hat{\beta}$ is the value of β that maximizes the log-likelihood function $\ell(\beta)$. Unlike ordinary least squares, this maximizer cannot be written down in closed form. Instead, it must be computed numerically.

The most common approach is Newton's method. To understand the idea, suppose we wish to maximize a smooth function $f(x)$ of a single variable. Starting from a current guess x_0 , we approximate f near x_0 by its second-order Taylor expansion

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0) + \frac{1}{2}f''(x_0)(x - x_0)^2.$$

Thus, near x_0 , the function is approximated by a quadratic. Since a quadratic can be optimized explicitly, we maximize this approximation instead of the original function.

Differentiating the quadratic approximation and setting the derivative equal to zero gives

$$f'(x_0) + f''(x_0)(x - x_0) = 0.$$

Solving for x yields the update

$$x = x_0 - \frac{f'(x_0)}{f''(x_0)}.$$

This becomes the next iterate,

$$x_{r+1} = x_r - \frac{f'(x_r)}{f''(x_r)}.$$

In several dimensions exactly the same idea applies. At the current iterate $\beta^{(r)}$, we replace the log-likelihood by its quadratic Taylor approximation and maximize that quadratic.

More precisely, if β_0 is a point in parameter space, then the second-order Taylor approximation of $\ell(\beta)$ around β_0 is

$$\ell(\beta) \approx \ell(\beta_0) + \nabla \ell(\beta_0)^T (\beta - \beta_0) + \frac{1}{2}(\beta - \beta_0)^T H\ell(\beta_0)(\beta - \beta_0),$$

where $\nabla \ell(\beta_0)$ denotes the gradient vector and $H\ell(\beta_0)$ denotes the Hessian matrix.

To maximize this quadratic approximation, we differentiate with respect to β . Using the fact that

$$\nabla_{\beta} \{(\beta - \beta_0)^T H\ell(\beta_0)(\beta - \beta_0)\} = 2H\ell(\beta_0)(\beta - \beta_0),$$

the gradient of the quadratic approximation is

$$\nabla\ell(\beta_0) + H\ell(\beta_0)(\beta - \beta_0).$$

Setting this equal to zero yields

$$\nabla\ell(\beta_0) + H\ell(\beta_0)(\beta - \beta_0) = 0.$$

Solving for β gives

$$\beta = \beta_0 - (H\ell(\beta_0))^{-1} \nabla\ell(\beta_0).$$

This point is the maximizer of the quadratic approximation. Replacing β_0 by the current iterate $\beta^{(r)}$ produces the updates in Newton's method:

$$\beta^{(r+1)} = \beta^{(r)} - (H\ell(\beta^{(r)}))^{-1} \nabla\ell(\beta^{(r)}), \quad (5)$$

where

$$\nabla\ell(\beta) = \left(\frac{\partial\ell}{\partial\beta_0}, \dots, \frac{\partial\ell}{\partial\beta_m} \right)^T$$

is the gradient vector and

$$H\ell(\beta) = \left(\frac{\partial^2\ell}{\partial\beta_i\partial\beta_j} \right)_{i,j}$$

is the Hessian matrix.

Geometrically, Newton's method repeatedly replaces the log-likelihood by a quadratic approximation, computes the maximizer of that quadratic, and then moves to the resulting point. Near the true maximizer, the approximation is extremely accurate, and Newton's method converges very rapidly. Under suitable smoothness assumptions, Newton's method converges quadratically once the iterates are sufficiently close to the optimum; see Boyd and Vandenberghe [2]. This means that if

$$e_r = \|\beta^{(r)} - \beta^*\|$$

denotes the error at iteration r , then

$$e_{r+1} \leq C e_r^2$$

for some constant $C > 0$. Consequently, the number of correct digits roughly doubles at each iteration.

Now let us explicitly write out the Newton updates. As we saw in (2),

$$\nabla \ell(\beta) = \sum_{i=1}^n (y_i - \pi_i) x_i,$$

where

$$\pi_i = \frac{\exp(x_i^T \beta)}{1 + \exp(x_i^T \beta)}.$$

Let π be the $n \times 1$ vector with entries $\pi_1, \dots, \pi_n$. These are the fitted probabilities at β .

Then

$$\nabla \ell(\beta) = X^T (y - \pi).$$

Also from (3),

$$H\ell(\beta) = -X^T W X,$$

where W is the diagonal matrix with entries

$$W_{ii} = \pi_i(1 - \pi_i).$$

Therefore Newton's update becomes

$$\beta^{(r+1)} = \beta^{(r)} + (X^T W X)^{-1} X^T (y - \pi).$$

This can be rewritten as

$$\beta^{(r+1)} = (X^T W X)^{-1} X^T W Z, \tag{6}$$

where

$$Z = X\beta^{(r)} + W^{-1}(y - \pi). \tag{7}$$

2.3.1 Iterative Reweighted Least Squares

The form of (6) should look familiar. Recall that in weighted least squares, given a response vector z , design matrix X , and positive definite weight matrix W , one minimizes

$$(z - X\beta)^T W (z - X\beta).$$

Differentiating with respect to β and setting the derivative equal to zero gives the normal equations

$$X^T W X \beta = X^T W z.$$

Provided $X^T W X$ is invertible, the solution is

$$\hat{\beta} = (X^T W X)^{-1} X^T W z.$$

Comparing this expression with (6), we see that the Newton update $\beta^{(r+1)}$ is exactly the weighted least squares estimator obtained by using

$$z = Z$$

as the response and

$$W = \text{diag}(\pi_1(1 - \pi_1), \dots, \pi_n(1 - \pi_n))$$

as the weight matrix.

Thus a single Newton step can be computed by solving a weighted least squares problem. After computing a new estimate $\beta^{(r)}$, we must recompute the fitted probabilities π_i , construct a new weight matrix W , construct a new pseudo-response vector Z , and solve another weighted least squares problem.

In other words, logistic regression is solved by repeatedly fitting weighted least squares models, with the weights changing from one iteration to the next. For this reason the algorithm is known as *iteratively reweighted least squares* (IRLS).

The weights have a natural interpretation. Since

$$W_{ii} = \pi_i(1 - \pi_i),$$

observations whose fitted probabilities are near 1/2 receive the largest weights, while observations whose fitted probabilities are very close to 0 or 1 receive relatively small weights. The algorithm therefore pays the most attention to observations near the current classification boundary, where the uncertainty is greatest.

3 A Different Modelling Approach to Classification

In logistic regression, we modeled the conditional probabilities

$$P(Y = k \mid X = x) \tag{8}$$

directly. This is a natural strategy because these probabilities are precisely the quantities needed for classification at a new test point.

There is, however, another possibility. Instead of modeling the conditional distribution of the class label given the predictors, we can model how the predictors are generated within each class. More precisely, we can model the class probabilities

$$P(Y = k) \tag{9}$$

together with the class-conditional densities

$$p(x \mid Y = k). \tag{10}$$

For discriminant analysis, the conditional class probabilities are written using Bayes' theorem as

$$\Pr(Y = k \mid X = x) = \frac{\pi_k f_k(x)}{\sum_{l=1}^K \pi_l f_l(x)}, \quad k = 1, \dots, K, \tag{11}$$

where

- $f_k(x) = p(x \mid Y = k)$ is the class-conditional density of the predictor vector X in class k . In LDA and QDA, these densities are modeled as multivariate normal distributions, separately for each class.
- $\pi_k = \Pr(Y = k)$ is the prior probability (or marginal probability) of class k .

Thus, discriminant analysis proceeds by modeling the class-conditional densities f_k and the class priors π_k , and then uses Bayes' theorem to obtain the posterior probabilities $\Pr(Y = k \mid X = x)$. Methods based on this idea are called *generative classifiers*. The most important examples are Linear Discriminant Analysis (LDA), Quadratic Discriminant Analysis (QDA), and Naive Bayes.

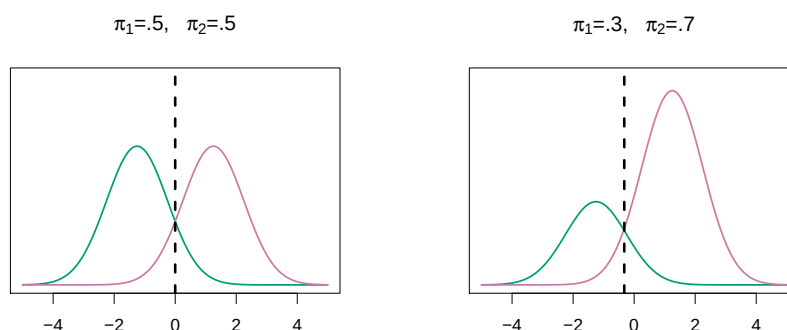
To classify an observation with feature vector x , we assign it to the class with the largest posterior probability:

$$\hat{C}(x) = \arg \max_k P(Y = k \mid X = x).$$

Since the denominator in Bayes' formula is the same for every class, this is equivalent to assigning x to the class that maximizes

$$\pi_k f_k(x).$$

Classify to the highest density



We classify a new point according to which density is highest.

When the priors are different, we take them into account as well, and compare $\pi_k f_k(x)$. On the right, we favor the pink class — the decision boundary has shifted to the left.

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3.1 Linear Discriminant Analysis

Linear discriminant analysis, or LDA, makes a specific modeling assumption. It assumes that the feature vector is normally distributed within each class:

$$X \mid Y = k \sim N(\mu_k, \Sigma).$$

The important point is that the means may differ across classes, but the covariance matrix is assumed to be the same for all classes.

Thus each class has its own center μ_k , but all classes have the same shape and orientation.

Why discriminant analysis?

- When the classes are well-separated, the parameter estimates for the logistic regression model are surprisingly unstable. Linear discriminant analysis does not suffer from this problem.
- If n is small and the distribution of the predictors X is approximately normal in each of the classes, the linear discriminant model is again more stable than the logistic regression model.
- Linear discriminant analysis is popular when we have more than two response classes, because it also provides low-dimensional views of the data.

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Figure 8: Some motivations for discriminant analysis. LDA can be more stable than logistic regression when the classes are well separated or when the sample size is small.

For a multivariate normal density,

$$f_k(x) = \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} \exp \left\{ -\frac{1}{2} (x - \mu_k)^T \Sigma^{-1} (x - \mu_k) \right\}.$$

To classify, we maximize

$$\pi_k f_k(x).$$

Equivalently, we maximize its logarithm:

$$\log \pi_k + \log f_k(x).$$

Ignoring terms that do not depend on k , this leads to the discriminant score

$$\delta_k(x) = x^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \log \pi_k.$$

The LDA classifier assigns x to the class with the largest discriminant score:

$$\hat{C}(x) = \arg \max_k \delta_k(x).$$

The key observation is that $\delta_k(x)$ is a linear function of x . Therefore the decision boundaries between classes are linear. This is why the method is called *linear* discriminant analysis.

3.2 Estimating the Parameters in LDA

Suppose we observe training data

$$(x_1, y_1), \dots, (x_n, y_n),$$

where $y_i \in \{1, \dots, K\}$. Under the LDA model,

$$\Pr(Y = k) = \pi_k, \quad X \mid Y = k \sim N(\mu_k, \Sigma),$$

with a common covariance matrix Σ shared across all classes.

The joint likelihood is

$$L(\pi, \mu, \Sigma) = \prod_{i=1}^n \pi_{y_i} \phi_p(x_i; \mu_{y_i}, \Sigma),$$

where $\phi_p(\cdot; \mu, \Sigma)$ denotes the p -dimensional Gaussian density.

Maximizing this likelihood yields the estimators

$$\hat{\pi}_k = \frac{n_k}{n},$$

$$\hat{\mu}_k = \frac{1}{n_k} \sum_{i:y_i=k} x_i,$$

and

$$\hat{\Sigma} = \frac{1}{n} \sum_{k=1}^K \sum_{i:y_i=k} (x_i - \hat{\mu}_k)(x_i - \hat{\mu}_k)^T.$$

(One often replaces $1/n$ by $1/(n - K)$ to obtain an unbiased estimate of Σ .)

The fitted discriminant score is

$$\hat{\delta}_k(x) = x^T \hat{\Sigma}^{-1} \hat{\mu}_k - \frac{1}{2} \hat{\mu}_k^T \hat{\Sigma}^{-1} \hat{\mu}_k + \log \hat{\pi}_k,$$

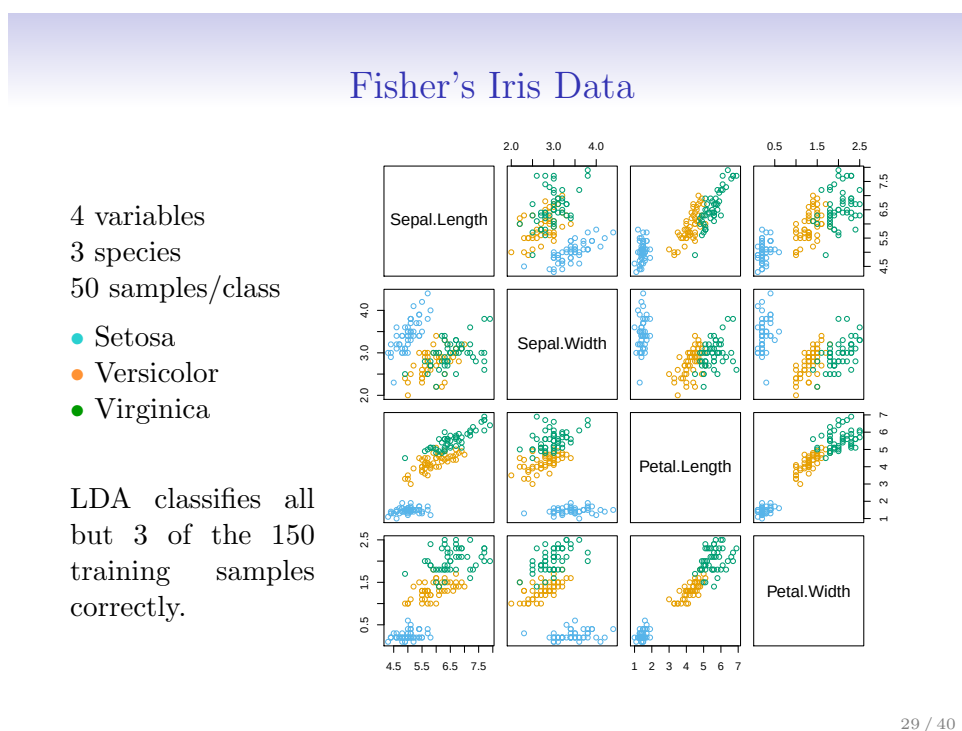
and the LDA classifier assigns x to the class

$$\hat{C}(x) = \arg \max_k \hat{\delta}_k(x).$$

The discriminant scores can be converted back into estimated class probabilities since $P(Y = k \mid X = x)$ is proportional to $\hat{\delta}_k(x)$.

Historically, this method was used by Fisher in the iris data set which is a classical example in classification. It contains measurements on three species of iris flowers:

setosa, versicolor, virginica.



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Figure 9: Fisher's iris data. The goal is to classify flowers into species using their measured features.

3.3 Quadratic Discriminant Analysis

The assumption of a common covariance matrix may be too restrictive. In many problems, different classes have different shapes, spreads, or orientations.

Quadratic discriminant analysis, or QDA, relaxes this assumption. It assumes

$$X \mid Y = k \sim N(\mu_k, \Sigma_k),$$

where each class has its own covariance matrix Σ_k .

The class-conditional density is now

$$f_k(x) = \frac{1}{(2\pi)^{p/2}|\Sigma_k|^{1/2}} \exp \left\{ -\frac{1}{2}(x - \mu_k)^T \Sigma_k^{-1} (x - \mu_k) \right\}.$$

Taking logs and dropping terms that do not depend on x , the QDA discriminant score is

$$\delta_k(x) = -\frac{1}{2} \log |\Sigma_k| - \frac{1}{2}(x - \mu_k)^T \Sigma_k^{-1} (x - \mu_k) + \log \pi_k.$$

The classifier is again

$$\hat{C}(x) = \arg \max_k \hat{\delta}_k(x).$$

But now $\delta_k(x)$ is a quadratic function of x . Therefore the decision boundaries are quadratic curves or surfaces.

Other forms of Discriminant Analysis

$$\Pr(Y = k | X = x) = \frac{\pi_k f_k(x)}{\sum_{l=1}^K \pi_l f_l(x)}$$

When $f_k(x)$ are Gaussian densities, with the same covariance matrix Σ in each class, this leads to linear discriminant analysis. By altering the forms for $f_k(x)$, we get different classifiers.

- With Gaussians but different Σ_k in each class, we get *quadratic discriminant analysis*.
- With $f_k(x) = \prod_{j=1}^p f_{jk}(x_j)$ (conditional independence model) in each class we get *naïve Bayes*. For Gaussian this means the Σ_k are diagonal.
- Many other forms, by proposing specific density models for $f_k(x)$, including nonparametric approaches.

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Figure 10: Other forms of discriminant analysis. Allowing a different covariance matrix in each class leads to QDA.

LDA is more restrictive but estimates fewer parameters. QDA is more flexible but requires estimating a separate covariance matrix for each class. Therefore QDA may perform better when the Gaussian model is approximately correct and there is enough data, while LDA may be more stable when the sample size is small.

Naive Bayes

Assumes features are independent in each class.

Useful when p is large, and so multivariate methods like QDA and even LDA break down.

- Gaussian naive Bayes assumes each Σ_k is diagonal:

$$\begin{aligned}\delta_k(x) &\propto \log \left[\pi_k \prod_{j=1}^p f_{kj}(x_j) \right] \\ &= -\frac{1}{2} \sum_{j=1}^p \left[\frac{(x_j - \mu_{kj})^2}{\sigma_{kj}^2} + \log \sigma_{kj}^2 \right] + \log \pi_k\end{aligned}$$

- can use for *mixed* feature vectors (qualitative and quantitative). If X_j is qualitative, replace $f_{kj}(x_j)$ with probability mass function (histogram) over discrete categories.

Despite strong assumptions, naive Bayes often produces good classification results.

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Figure 11: Naive Bayes assumes that the features are conditionally independent within each class. This can be useful when the number of features is large.

The conditional independence assumption is often unrealistic. For example, in a spam filtering problem, the appearances of different words in an email are not really independent. Nevertheless, Naive Bayes can work surprisingly well in high-dimensional problems because it avoids estimating a large covariance matrix. Perhaps we do not need to estimate the densities very well as long as we can classify well.

3.4 Logistic Regression versus LDA

Logistic Regression versus LDA

For a two-class problem, one can show that for LDA

$$\log \left(\frac{p_1(x)}{1 - p_1(x)} \right) = \log \left(\frac{p_1(x)}{p_2(x)} \right) = c_0 + c_1 x_1 + \dots + c_p x_p$$

So it has the same form as logistic regression.

The difference is in how the parameters are estimated.

- Logistic regression uses the conditional likelihood based on $\Pr(Y|X)$ (known as *discriminative learning*).
- LDA uses the full likelihood based on $\Pr(X, Y)$ (known as *generative learning*).
- Despite these differences, in practice the results are often very similar.

Footnote: logistic regression can also fit quadratic boundaries like QDA, by explicitly including quadratic terms in the model.

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Figure 12: Logistic regression and LDA both lead to linear log-odds in the two-class case, but they arrive there through different modeling assumptions.

	Logistic regression	LDA
Modeling approach	Models $P(Y X)$	Models $P(X Y)$
Type	Discriminative	Generative
Assumption on X	No Gaussian assumption	Gaussian class densities
Decision boundary	Linear	Linear
Fitting	Numerical optimization	Closed-form estimates
Small samples	Can be less stable	Can be more stable if assumptions hold

In large samples, logistic regression is often preferred because it makes fewer assumptions about the distribution of the predictors. However, when the Gaussian class-conditional model is approximately correct, LDA can be very efficient and stable. Modern machine learning contains descendants of both viewpoints. Deep neural networks used for classification are typically discriminative models, while methods such as Gaussian mixture models, hidden Markov models, variational autoencoders, and diffusion models continue the generative modeling tradition.

4 Imbalanced Classes

In many classification problems, one class occurs much less frequently than the others. Examples include credit card default, fraud detection, rare diseases etc. Such problems are often called *imbalanced classification problems*.

To see why imbalance matters, suppose that only 2% of customers default on a loan. A classifier that predicts “no default” for every customer will achieve an accuracy of approximately 98%, despite being completely useless for identifying defaulters.

This illustrates an important lesson: classification accuracy alone can be highly misleading when classes are imbalanced.

4.1 The Confusion Matrix

For a binary classification problem, predictions are summarized by the confusion matrix (it is indeed very confusing!)

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

Several useful performance measures are derived from these quantities.

The *sensitivity* (or true positive rate) is

$$\text{Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad (12)$$

while the *specificity* (or true negative rate) is

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}}. \quad (13)$$

Sensitivity measures the proportion of positive cases that are correctly identified, whereas specificity measures the proportion of negative cases that are correctly classified.

4.2 Choosing a Classification Threshold

Logistic regression produces estimated probabilities

$$\hat{p}(x) = P(Y = 1 \mid X = x). \quad (14)$$

To obtain a classifier, we choose a threshold t and predict

$$\hat{Y} = \begin{cases} 1, & \hat{p}(x) > t, \\ 0, & \hat{p}(x) \leq t. \end{cases} \quad (15)$$

The default choice $t = 0.5$ is often convenient but is not always appropriate. In medical screening, for example, failing to detect a disease may be much more costly than raising a false alarm. In such situations one may deliberately choose a smaller threshold in order to increase sensitivity.

4.3 ROC Curves

4.3.1 The Population ROC Curve

The setting is that the pair (X, Y) has a joint distribution and Y is binary (positive means $Y = 1$). Suppose a classifier produces a real-valued score $S(X)$ where larger values indicate stronger evidence that an observation belongs to the positive class.

For example, in logistic regression a natural choice is

$$S(X) = \hat{p}(X), \quad (16)$$

the estimated probability that the observation belongs to the positive class. More generally, any real-valued score may be used. At this stage we are not yet classifying observations. We are simply assigning a score to each observation.

To understand the quality of a scoring rule, it is useful to consider the distribution of scores among positive and negative observations separately. Let

$$S^+ \sim \text{distribution of } S(X) \mid Y = 1, \quad (17)$$

and

$$S^- \sim \text{distribution of } S(X) \mid Y = 0. \quad (18)$$

A good classifier should assign larger scores to positive observations than to negative observations. Thus we would like the distribution of S^+ to lie mostly to the right of the distribution of S^- .

To obtain an actual classifier, we must choose a threshold t and classify according to

$$\hat{Y} = \begin{cases} 1, & S(X) \geq t, \\ 0, & S(X) < t. \end{cases} \quad (19)$$

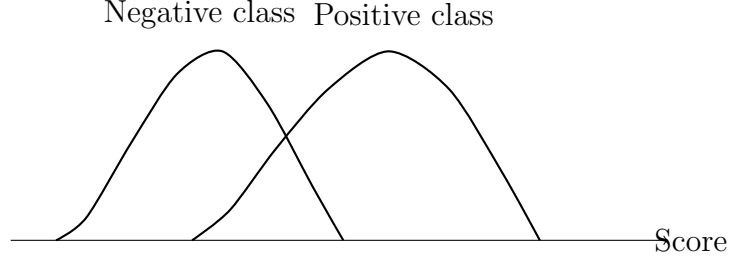


Figure 13: The ROC curve and AUC are determined by the degree of overlap between the score distributions of the two classes. Less overlap corresponds to better classification performance.

For this threshold, the true positive rate is

$$\text{TPR}(t) = P(S^+ \geq t), \quad (20)$$

and the false positive rate is

$$\text{FPR}(t) = P(S^- \geq t). \quad (21)$$

Notice that both quantities are population probabilities. No dataset has yet been introduced.

As the threshold t varies over all possible values, we obtain a collection of points

$$(\text{FPR}(t), \text{TPR}(t)). \quad (22)$$

The *population ROC curve* for the score S is the set of all such points:

$$\text{ROC} = \{ (P(S^- \geq t), P(S^+ \geq t)) : t \in \mathbb{R} \}. \quad (23)$$

Thus the ROC curve summarizes how the distributions of S^+ and S^- compare across all possible thresholds. The shape of the ROC curve reflects the degree of separation between the two score distributions.

- If the distributions of S^+ and S^- are nearly identical, then

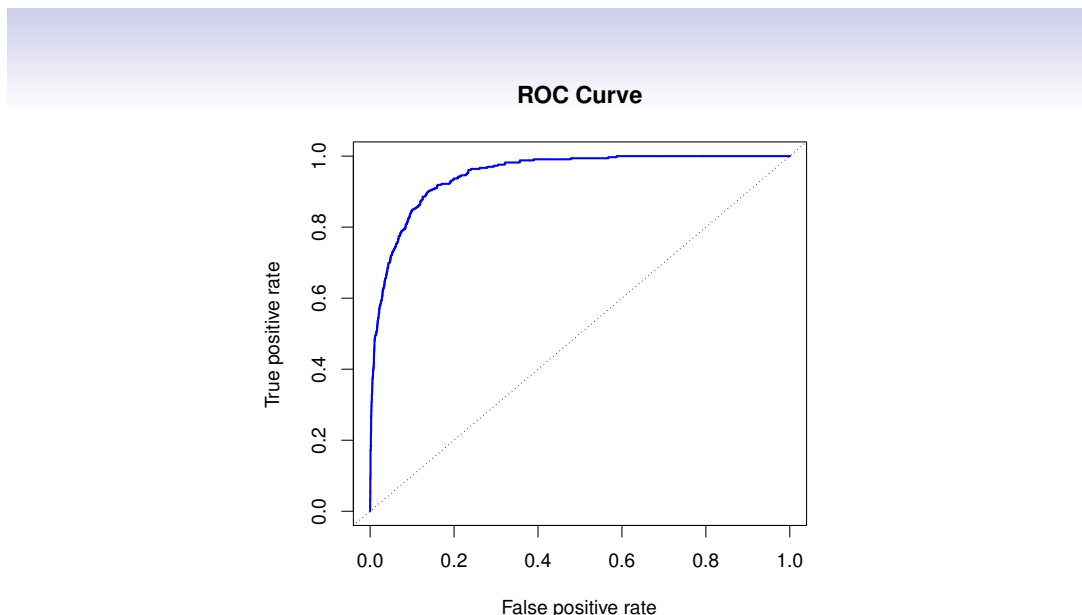
$$P(S^+ \geq t) \approx P(S^- \geq t) \quad (24)$$

for every threshold t . The ROC curve is then close to the diagonal line

$$y = x, \quad (25)$$

which corresponds to a classifier with little discriminative power.

- If positive observations almost always receive larger scores than negative observations, then the ROC curve hugs the left and upper boundaries of the unit square. In this case the classifier is nearly perfect.



The *ROC plot* displays both simultaneously.

Sometimes we use the *AUC* or *area under the curve* to summarize the overall performance. Higher *AUC* is good.

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4.4 The Area Under the ROC Curve

The ROC curve summarizes the performance of a scoring rule across all possible thresholds. A convenient way to summarize the entire curve by a single number is through the *area under the ROC curve*, commonly abbreviated as *AUC*.

At the population level, the AUC is simply the area under the population ROC curve. Since the ROC curve lies inside the unit square, we always have

$$0 \leq \text{AUC} \leq 1. \quad (26)$$

A classifier with no discriminative power has an ROC curve close to the diagonal line ($y=x$), and hence an AUC close to $\frac{1}{2}$.

A perfect classifier has an ROC curve that passes through the upper-left corner of the unit square, giving $\text{AUC} = 1$.

Thus larger values of the AUC correspond to better separation between the positive and negative classes.

The AUC has a particularly elegant probabilistic interpretation. Recall the population score distributions introduced in the previous section.

The AUC has a particularly elegant probabilistic interpretation. Let

$$S_1 \sim \mathcal{L}(S(X) \mid Y = 1), \quad S_0 \sim \mathcal{L}(S(X) \mid Y = 0),$$

be independent random variables having the score distributions of the positive and negative classes respectively.

One can show that

$$\text{AUC} = P(S_1 > S_0),$$

provided ties occur with probability zero (**Exercise!**). More generally,

$$\text{AUC} = P(S_1 > S_0) + \frac{1}{2}P(S_1 = S_0).$$

Thus the AUC is the probability that if we independently draw one positive observation and one negative observation from the population, the positive observation receives the larger score.

4.4.1 Sample ROC Curve

In practice, the distributions of the scores in the positive and negative classes are unknown. Suppose a classifier has been trained using a training dataset and produces a score $S(X)$ for each observation. We then evaluate the classifier on an independent test dataset. For any threshold t , we estimate

$$P(S(X) \geq t \mid Y = 1)$$

and

$$P(S(X) \geq t \mid Y = 0)$$

by the corresponding sample proportions among the positive and negative observations in the test data. These estimates give the empirical true positive rate and false positive rate.

Each threshold produces one point given by the empirical false positive rate and true positive rate (FPR, TPR). Varying the threshold over all possible values traces out the *empirical ROC curve*, which may be viewed as a sample-based estimate of the population ROC curve. The AUC provides a convenient way to compare different classification methods that produce scores or estimated probabilities.

Remark 4.1. *Importantly, the ROC curve and the AUC depend only on the ordering of the scores assigned to observations. Any strictly increasing transformation of the scores leaves the ROC curve and the AUC unchanged. Thus the AUC measures how well the scoring rule ranks positive observations ahead of negative observations, rather than the accuracy of the numerical probabilities themselves.*

To summarize, when classes are highly imbalanced, the classification threshold should be chosen carefully and sensitivity, specificity, and AUC often provide more informative measures of performance.

5 Generalized Linear Models

Linear regression and logistic regression both belong to a broader family known as *generalized linear models* (GLMs). The GLM framework provides a unified way to model a wide variety of response variables and is one of the classical ideas in statistics [3].

In linear regression, we model the conditional mean directly:

$$E(Y_i | X_i = x_i) = x_i^T \beta. \quad (27)$$

For a binary response, with $p_i = E(Y_i | X_i = x_i)$, instead of modeling the probability directly, logistic regression models the log-odds:

$$\log\left(\frac{p_i}{1 - p_i}\right) = x_i^T \beta. \quad (28)$$

This transformation automatically guarantees that the fitted probabilities remain between 0 and 1.

This idea of using some transformation of the mean response and then modelling it as a linear function of X is useful more generally and gives rise to GLM. A generalized linear model consists of two ingredients:

1. A probability distribution for the response variable, like Gaussian, Bernoulli, Poisson etc.
2. A link function applied to the conditional mean of the response which is then modelled as a linear function of X .

If $\mu_i = E(Y_i | X_i = x_i)$ then the general GLM form is

$$g(\mu_i) = x_i^T \beta, \quad (29)$$

where g is called the *link function*.

5.1 Poisson Regression

Many response variables represent counts rather than continuous measurements. Examples include the number of bicycles rented in an hour, the number of insurance claims filed by a customer, the number of visits to a website, and the number of hospital admissions in a day. Such data are often modeled using the Poisson distribution.

A random variable Y is said to have a Poisson distribution with parameter $\lambda > 0$ if

$$P(Y = k) = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k = 0, 1, 2, \dots \quad (30)$$

The parameter λ is both the mean and the variance,

$$E(Y) = \lambda, \quad \text{Var}(Y) = \lambda. \quad (31)$$

As a concrete example, suppose Y_i denotes the number of bicycles rented during a particular hour. We might expect the average number of rentals to depend on variables such as temperature, season, weather conditions, and the time of day. Since the response consists of nonnegative counts, ordinary linear regression is not ideal because it can produce negative fitted values. Also, it is possible that the variance of Y is not the same for all X (heteroscedastic), if we think that the variances are roughly equal to the mean, along with Y being nonnegative integer valued, then modelling Y as a Poisson random variable becomes natural.

Poisson regression assumes that

$$Y_i | X_i = x_i \sim \text{Poisson}(\lambda_i). \quad (32)$$

Since λ_i must be positive, we model its logarithm as a linear function of the predictors:

$$\log(\lambda_i) = x_i^T \beta. \quad (33)$$

Equivalently,

$$\lambda_i = \exp(x_i^T \beta). \quad (34)$$

This guarantees that the fitted mean is always positive.

5.2 Interpreting the Coefficients

Suppose predictor x_j increases by one unit while all other predictors remain fixed.

Then

$$\log(\lambda) \longrightarrow \log(\lambda) + \beta_j, \quad (35)$$

which implies

$$\lambda \longrightarrow e^{\beta_j} \lambda. \quad (36)$$

Thus e^{β_j} represents the multiplicative change in the expected count associated with a one-unit increase in predictor j . Again we really need to interpret this as the multiplicative change adjusted for all the other predictors. This multiplicative interpretation is often more natural for count data than the additive interpretation encountered in ordinary linear regression.

5.3 Maximum Likelihood Estimation

For independent observations, the likelihood is

$$L(\beta) = \prod_{i=1}^n e^{-\lambda_i} \frac{\lambda_i^{y_i}}{y_i!}, \quad \lambda_i = \exp(x_i^T \beta). \quad (37)$$

As with logistic regression, the maximum likelihood estimator cannot be written in closed form and is computed numerically using Newton's method.

5.4 Other Examples of GLMs

Response Type	Distribution	Typical Mean Constraint	Common Link
Continuous measurements	Gaussian	$-\infty < \mu < \infty$	Identity
Binary outcomes	Bernoulli	$0 < p < 1$	Logit
Count data	Poisson	$\lambda > 0$	Log
Positive continuous responses	Gamma	$\mu > 0$	Log
Overdispersed counts	Negative Binomial	$\mu > 0$	Log

Gamma regression is frequently used for positive continuous quantities such as insurance costs, waiting times, and rainfall amounts. Negative binomial regression is commonly used for count data when the variance substantially exceeds the mean, a phenomenon known as *overdispersion*.

5.5 Summary

Linear regression models a conditional mean, logistic regression models a probability, and Poisson regression models a count rate. Generalized linear models reveal that all three are examples of the same underlying idea:

Choose an appropriate probability model for the response, transform its mean using a suitable link function, and then model the transformed mean as a linear function of the predictors.

Remark 5.1. *The examples discussed in this chapter are instances of a much broader theory based on exponential-family distributions. One of the remarkable features of this theory is that many seemingly different statistical models can be studied using the same mathematical tools. In particular, maximum likelihood estimation, Newton’s method, iteratively reweighted least squares (IRLS), standard error calculations, and asymptotic normality can all be developed simultaneously for a large class of models. From this viewpoint, linear regression, logistic regression, and Poisson regression are not three different techniques but three instances of the same modelling principle. The choice of response distribution and link function determines the model, while estimation and inference are carried out using essentially the same ideas. This perspective was one of the major achievements of twentieth-century statistics and is one of the reasons generalized linear models remain so widely used today; see [1] for a modern introduction to GLM.*

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